What Missionary Control Left in the Language of Two 1880s isiXhosa Newspapers

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Abstract

From 1884 the missionary *Isigidimi sama-Xosa* and the African-owned *Imvo Zabantsundu* ran side by side, both with African editors. We ask whether missionary control shows in *what* they printed (salience) or in *how* they used words (framing). Isigidimi uses religious words about four times as often (exploratory), and Imvo parliamentary words 3.3 times as often (neither survives correction for 15 tests). The religious gap holds in readers' letters too, so it reflects who wrote to each paper, not only the editors. Framing evidence is weak: a religious lean of Isigidimi's political words rests on one word, *umhlaba*, which the papers use in different senses, and a neutral control leans further. Modern tokenizers charge 1.7–12% more for this text; old spelling alone penalises only the African-language model. Control shaped what the papers printed; whether it shaped how they wrote lies below our test's detection floor.

Introduction and Background

Where did missionary control leave its trace in the language of the earliest isiXhosa press? The IsiXhosa Intellectual Traditions archive (Schoots et al., 2023a) makes this testable. *Isigidimi sama-Xosa* was a missionary paper; *Imvo Zabantsundu*, founded in 1884 by Isigidimi's former editor J. T. Jabavu, was the first African-owned isiXhosa paper, free "from any missionary control" (Schoots et al., 2023a, p. 55). From late 1884 both had African editors, Gqoba and Jabavu (Schoots, 2023b, p. 71), so comparing them isolates *control*, not the editor's identity.

Communication research separates *agenda-setting*, what a paper covers (McCombs & Shaw, 1972), from *framing*, how it presents it (Entman, 1993). We measure the first as **salience**, how often a topic's words appear, and the second with word embeddings, which place a word by the company it keeps (Hamilton et al., 2016a). Such studies mostly use modern, high-resource news (Card et al., 2015; Field et al., 2018); for these papers, Schoots (2023b) counted root words over time. We add controls that make both kinds of difference testable, a split into readers' letters and the rest to ask *how* control worked, and an audit of what this text costs modern tokenizers (Petrov et al., 2023; Rust et al., 2021). **H1**: the papers differ in how often they use political and religious words. **H2**: Isigidimi's political words sit closer to religious contexts. **H3**: historical text and its spelling raise tokenizer cost.

Methods

Data. We use the archive's human-corrected transcriptions, which were not spell-checked (Schoots et al., 2023a, p. 59), from 1 November 1884 to 1888 (Isigidimi's 1886 editions are missing): 286,663 tokens of Isigidimi and 1.31M of Imvo.

Threat	Response
Papers cover different years	Only Nov 1884–1888; Imvo pages drawn month by month
Imvo is 4.6× larger	Ten Imvo samples of Isigidimi's size
Frequency distorts comparisons	Rank results among 2000 frequency-matched random word sets (cf. Monroe et al., 2008)
Papers change over time	Measure each paper's own drift in other years
Embeddings vary by run	Tune for stability (Antoniak & Mimno, 2018); average ten runs
Effects may not be political	Four neutral control concepts

Table 1: Threats to a fair comparison, and our response.

Salience. Concepts are word stems adapted from Schoots (2023b) (Table S1): ten political concepts fixed in advance, four neutral controls, and religion (*tixo*, *kristu*). Religion is exploratory: the archive's authors expect it (Schoots et al., 2023a, p. 57), but a classifier trained on the same years selected it. Ratios carry issue-bootstrap CIs.

Letters. A letter opens by addressing the editor (*nkosi mhleli … kaundifakele*, 'sir, editor … please print') and ends with a signature. This rule found 28 of 30 genuine letters in a random hand check (Table S5); letters are 6% of Isigidimi and 7% of Imvo.

Framing. We train fastText embeddings (Bojanowski et al., 2017) (window 1, 400 dimensions; more stable than word2vec). A word's *profile shift* is $1 - r$ between its similarities to 500 common words in each paper (adapted from Hamilton et al., 2016b), and a concept's shift is the mean over its forms; *neighbour overlap* compares its ten nearest neighbours (Gonen et al., 2020). Chance comes from shuffling issues between papers (Dubossarsky et al., 2017). A directional test asks whether political words sit nearer religious or political anchor words (Table S4). To read it, we count religious (heaven, God) and property or political words (title, law) within ten words of *umhlaba*.

Tokenizers. We count subwords for XLM-R (Conneau et al., 2020), Serengeti (Adebara et al., 2023) and mBERT on modern news (Adelani et al., 2023), on the same news in old spelling (per word) and on our text (per character, as word division differs).

Results

H1: the papers print different things. Isigidimi uses religious words 4.2 times as often as Imvo ($p = 0.020$ against matched sets) and Imvo parliamentary words 3.3 times as often ($p = 0.018$; Fig. 1). No other concept differs beyond chance; neither gap survives correction for 15 tests ($p < 0.003$). The religious gap exceeds each paper's own drift between eras; the parliamentary gap may not (Isigidimi drifts 1.8×, CI to 5.0; Table S3). The religious gap holds in letters (Imvo/Isigidimi

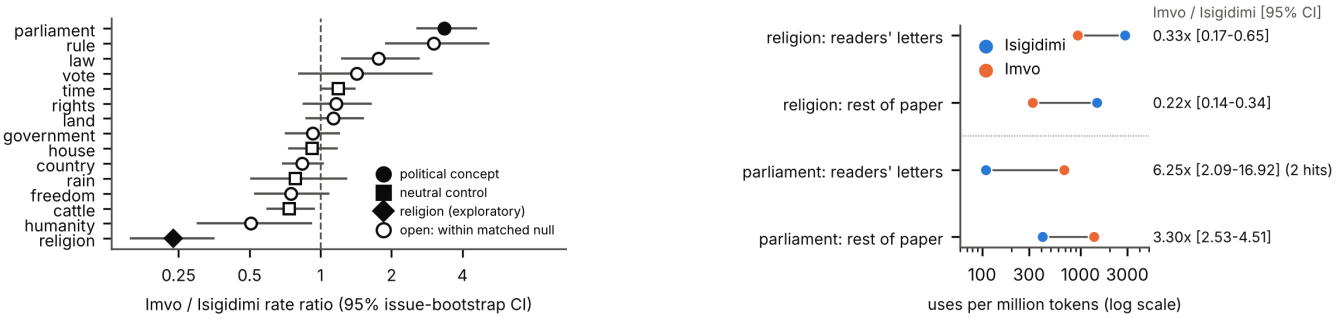


Figure 1: What the papers printed. Left: Imvo/Isigidimi rate ratio (95% CI); filled: $p < 0.05$ against matched sets, uncorrected. Right: the religious gap holds in readers' letters as in the rest of each paper.

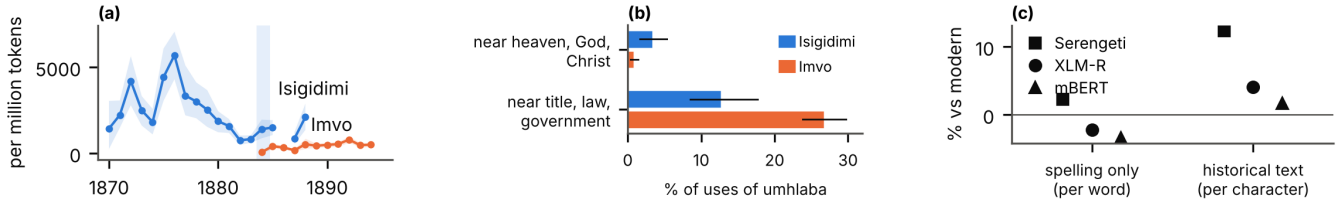


Figure 2: (a) Isigidimi's religious vocabulary was falling long before Imvo (light blue: twice-monthly issues, Schoots et al., 2023a, p. 59). (b) Isigidimi uses *umhlaba* more often near religious words, Imvo near property and political words. (c) Old spelling (per word) raises cost only for Serengeti; historical text (per character) for all.

0.33, CI 0.17–0.65) as in the rest (0.22); parliament is too rare in Isigidimi's letters to judge (2 uses).

H2: framing differences are weak and not specific. Word use differs overall (profile shift 0.348 against 0.230 by chance), but no concept stands out: parliament reaches the 96th percentile of matched sets, the control *cattle* the 90th–99th (Fig. S1). Neighbour overlap for parliament and vote shifts in three of five settings, profile shift in one. Political words lean towards religious anchors in Isigidimi in all five settings (Δ +0.44 to +0.76 z-units; Bonferroni-significant across the five in three; Table S2). But without *land* the lean is rarely significant ($p = 0.010$–0.256), the control *house* leans further (+0.84 to +1.51); dropping the anchor *endlu* ('in the house') leaves both leans in place (+0.62 and +1.04), and *umhlaba* sits near religious words in 3.4% of Isigidimi's uses (Imvo 0.9%) and near property or political words in 13% (Imvo 27%; Fig. 2b). A one-word concept must shift by 0.15 (1.645 matched-set SDs) to be detected, more than the overall gap (0.118).

H3: historical text costs more. Per character, our text needs 12.4%, 4.1% and 1.7% more subwords with Serengeti, XLM-R and mBERT. Old spelling alone raises Serengeti's cost by 2.2% per word but lowers XLM-R's and mBERT's (-2.2%, -3.4%; Fig. 2c).

Discussion

How control worked (H1). Control left a clear trace in what the papers printed, and not only through the editors: readers who wrote to Isigidimi used religious words three times as often as Imvo's, so the papers drew on, or chose, different writers. The gap is not a reaction to Imvo: Isigidimi's religious vocabulary had been falling since 1876, and rose again in 1884–85, around its change of editor and Imvo's arrival (Fig. 2a). Parliamentary words entered Isigidimi in 1883–84, late in Jabavu's editorship, when it appeared twice monthly and before the 1884 Cape election (Schoots, 2023b); these causes cannot be separated (Fig. S2).

What the embeddings see (H2). The framing signal looks like a difference in meaning, not slant. Isigidimi more often uses *umhlaba* for the earth of creation, Imvo for land under title and law; this alone would place the word differently in each model and could produce the political 'lean'. A neutral control behaves similarly, and narrow windows mostly capture grammar. With about 300k tokens per paper, we can rule out large framing differences, not small ones.

Responsible NLP (H3). Old spelling penalises only the model built for African languages, perhaps because Serengeti's vocabulary has twice XLM-R's share of *th*/*kh*/*ph* pieces (1.9% vs 0.9%); this is a one-model hypothesis, and Serengeti may have seen MasakhaNEWS in training. Normalising spelling is no fix: it would erase the orthography the archive preserves, irreversibly (old *t* may be modern *t* or *th*). The archive's proposed machine-learning labelling (Schoots et al., 2023a, p. 61) would inherit this cost. Our results describe two papers, not isiXhosa speakers; cultural readings without speaker review would be extractive (Bird, 2020).

Limitations and conclusion. We study two papers in one window, with a missing year, no editor metadata, a heuristic letter split, a modern-news domain mismatch and stems not yet reviewed by a first-language speaker (*rulumente* is 'government' here, 'parliament' in Schoots et al., 2023a, p. 62). Within these limits, control shaped what these papers printed, through their writers as well as their editors; whether it shaped how political language was used lies below our test's detection floor.

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Statement on the use of generative AI (Faculty of Science AI Ethics Statement)

Generative AI (Claude, by Anthropic, used through the Claude Code tool) helped to write the analysis code (data preparation, embedding experiments, statistics, figure scripts), to draft and edit the text and LaTeX source, and to fill in the checklist below from the course template. Generative AI was also used to review drafts and suggest revisions; suggested changes were checked against the code outputs and the cited sources before being kept. All numbers in the paper are produced by the submitted code and were read from its output files. The authors take full responsibility for all code and text submitted, including the accuracy of every claim and citation.

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    Answer: [Yes]

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    Justification: No crowdsourcing or human-subjects research was conducted. A first-language isiXhosa speaker is to be consulted as an expert adviser to check our interpretations; their judgments are not reported as data.

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Supplementary material

S1. Concepts, stems and the word forms they match (1 Nov 1884–1888, both papers)

Stems are matched as substrings of surface forms; following Schoots (2023b) we inspected forms for false positives (our first *law* stem also matched *inteto*, speech, and was dropped) but did not hand-filter them, so run-together forms (e.g. *rulumenteningomsebenzi*) and missed variants remain. All stems, forms and glosses await review by a first-language isiXhosa speaker. Political concepts were fixed before testing; religion was suggested by the classifier and is exploratory.

Concept	Type	Stems	Most frequent matched forms (count; number of distinct forms)
parliament	political	palamente	palamente (1754), ipalamente (47), palamenteni (22), epalamente (14), bapalamente (1), nepalamente (1), yepalamente (1); 11 forms
vote	political	voti	ivoti (301), voti (217), evotini (136), abavoti (98), ngevoti (55), levoti (22), nevoti (22); 115 forms
government	political	rulumente	rulumente (1882), rulumenteni (7), urulumente (5), serulumente (1), rulumenteningomsebenzi (1), zikarulumente (1), rulumenteayiko (1); 8 forms
rule	political	mbuso	zombuso (169), umbuso (76), mbuso (60), yombuso (42), kombuso (37), lombuso (28), kulombuso (23); 43 forms
law	political	mteto	umteto (724), lomteto (475), mteto (406), ngomteto (167), ngumteto (94), komteto (76), nomteto (76); 82 forms
rights	political	lungelo	amalungelo (315), ilungelo (288), lungelo (113), nelungelo (54), namalungelo (50), lilungelo (39), elilungelo (37); 115 forms
country	political	lizwe	lizwe (646), ilizwe (292), kwelilizwe (201), elilizwe (101), elizwe (96), belilizwe (76), yelilizwe (52); 78 forms
land	political	mhlaba	umhlaba (836), mhlaba (208), lomhlaba (156), nomhlaba (105), komhlaba (99), somhlaba (95), yomhlaba (94); 135 forms
freedom	political	kululek	inkululeko (27), kukululekile (24), ukululekile (13), ngenkululeko (10), abakulekileyo (8), nkululeko (8), nenkululeko (7); 92 forms
humanity	political	buntu	ubuntu (45), yobuntu (14), nobuntu (8), sobuntu (8), buntu (7), lobuntu (7), zobuntu (6); 29 forms
time	neutral	xesha	ixesha (787), ngexesha (364), xesha (298), amaxesha (243), lixesha (191), ngamaxesha (149), zamaxesha (114); 207 forms
house	neutral	ndlu	indlu (719), izindlu (328), kwindlu (276), ndlu (234), yendlu (227), ngendlu (133), nendlu (130); 193 forms
rain	neutral	mvula	imvula (267), mvula (31), yemvula (23), ezimvula (17), lemvula (16), yimvula (14), zimvula (10); 45 forms
cattle	neutral	nkomo	inkomo (555), zenkomo (258), nkomo (193), yenkomo (93), nenkomo (75), ngenkomo (35), lenkomo (33); 79 forms
religion	exploratory	tixo, kristu	tixo (684), kristu (207), makristu (7), kristukazi (7), mkristu (3), amakristu (2), lobukristu (1); 16 forms

S2. Secondary framing tests across settings

Percentile of the observed shift among 2000 frequency-matched word sets, and one-sided p (uncorrected). P+V = *palamente* + *voti*; P+V+R adds *rulumente* (uncertain meaning). Directional Δ: Isigidimi minus Imvo, religious minus political similarity (in z-score units), political concepts, two-sided p against matched sets; the last columns give the control *house*. Three settings survive Bonferroni over five ($p < 0.01$): window 1, window 2, and window 2 with forms joined. Joining partly disjunctive forms (*uku ba*) leaves the global profile gap unchanged. Matched sets are re-drawn for each test; without *land* the pooled set is smaller, so lower power may contribute to the higher p.

Setting	P+V profile	P+V Jaccard	P+V+R profile	P+V+R Jaccard	Δ	Δ p	p w/o land	house Δ	house p
window 1 (tuned)	92 (p=0.083)	96 (p=0.039)	92 (p=0.077)	99 (p=0.007)	+0.68	<0.001	0.010	+1.08	<0.001
window 1, forms joined	99 (p=0.010)	95 (p=0.053)	100 (p=0.001)	100 (p=0.002)	+0.44	0.016	0.088	+1.15	<0.001
window 2	86 (p=0.139)	94 (p=0.061)	90 (p=0.096)	99 (p=0.010)	+0.76	<0.001	0.045	+1.34	<0.001
window 2, forms joined	91 (p=0.094)	98 (p=0.023)	99 (p=0.011)	100 (p=0.002)	+0.58	0.009	0.064	+1.51	<0.001
window 5	62 (p=0.382)	98 (p=0.016)	81 (p=0.189)	100 (p=0.002)	+0.66	0.015	0.256	+0.84	0.061

S3. Era drift: does a concept's rate move across eras within one paper?

Per-million-token rates and ratios (later/earlier, or Imvo/Isigidimi) with issue-bootstrap 95% CIs.

Concept	Comparison	First	Second	Ratio (95% CI)
parliament	Isigidimi: later / earlier era	214	394	1.84 (0.96–4.98)
parliament	Imvo: later / earlier era	1320	758	0.57 (0.50–0.66)
parliament	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	214	758	3.54 (1.98–9.58)
government	Isigidimi: later / earlier era	1667	1088	0.65 (0.50–0.85)
government	Imvo: later / earlier era	1210	452	0.37 (0.31–0.45)
government	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	1667	452	0.27 (0.22–0.34)
vote	Isigidimi: later / earlier era	72	614	8.55 (3.51–18.94)
vote	Imvo: later / earlier era	799	921	1.15 (0.88–1.51)
vote	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	72	921	12.82 (7.85–25.92)
religion	Isigidimi: later / earlier era	2466	1556	0.63 (0.46–0.84)
religion	Imvo: later / earlier era	362	547	1.51 (1.15–2.03)
religion	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	2466	547	0.22 (0.18–0.28)

S4. Anchors for the directional test

Anchors are the 15 top-PMI neighbours of each seed pair in held-out text (Isigidimi 1876-83 + Imvo 1889-94), frozen before testing. Religious (15 words; seeds *tixo*, *kristu*): *sikelela*, *sindisa*, *ukumkanikazi*, *inkosazana*, *buka*, *oyingcwele*, *lilizwi*, *msindisi*, *nelizwi*, *kwelizwi*, *wanga*, *lika*, *yesu*, *izono*, *yehova*. Political (15 words; seeds *palamente*, *voti*): *emqulwini*, *wonyulo*, *lonyulo*, *elugqatsweni*, *eziqingata*, *lokunyula*, *kwesase*, *lokuvota*, *endlu*, *kwamagama*, *ukumela*, *izayo*, *ngalomteto*, *nawupina*, *kwabase*. Seed words cannot themselves be tested (Fig. S1). Political anchors include *endlu* ('in the house'), which might have favoured the neutral control *house*; re-running without *endlu* leaves both the political lean and *house*'s lean in place (outputs/framing_matched_tests_noendlu.csv), and *ngalomteto*, which contains the *law* stem. Glosses are unverified: two religious anchors (*ukumkanikazi*, *inkosazana*) may reflect royal rather than religious register, and several anchors on both sides (*lika*, *buka*; *nawupina*, *kwabase*, *izayo*) may be grammatical rather than topical, consistent with narrow-window PMI capturing grammar. All await speaker review.

S5. Readers' letters versus the rest of each paper

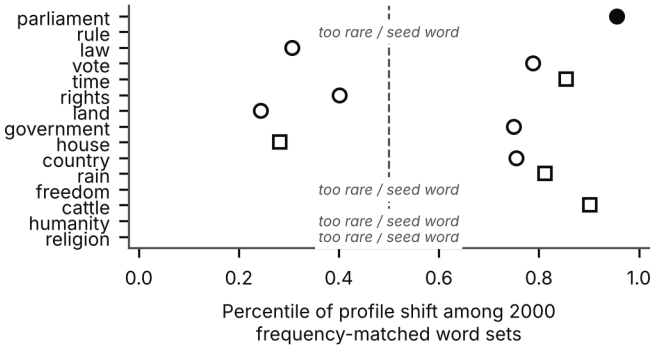
Letters are detected by src/12_letters.py: a letter opens by addressing the editor (vocative *mhleli* or *mcokeli*, not *u/ku/ngu mhleli*, which refer to him) or by asking for insertion (*ndifakele*, *kaundifakele*), and ends at the signature (a run of very short lines). A hand check of 30 random detections found 28 genuine letters; the errors were usifake umkala (horse advert: 'put a bit on it'); no mhleli wepepa lama bhulu ('and the editor of the Boer paper'). Recall was not measured. Imvo is restricted to the months in which Isigidimi has text; CIs from 4000 issue resamples; rates are not tested against matched sets. Hits = concept uses.

Part	Concept	Isig. tokens	Isig. hits	Imvo tokens	Imvo hits	Imvo/Isig.	95% CI
letters	parliament	18,344	2	70,475	48	6.25	2.09–16.92
letters	religion	18,344	52	70,475	66	0.33	0.17–0.65
letters	vote	18,344	4	70,475	58	3.77	1.04–18.93
letters	government	18,344	13	70,475	115	2.30	1.14–7.18
rest of paper	parliament	268,319	111	936,998	1280	3.30	2.53–4.51
rest of paper	religion	268,319	394	936,998	306	0.22	0.14–0.34
rest of paper	vote	268,319	172	936,998	821	1.37	0.79–2.83
rest of paper	government	268,319	299	936,998	902	0.86	0.67–1.13

S6. Which sense of *umhlaba*?

For every use of the land stem (*mhlaba*) we look at the 10 tokens either side. Religious cues: *izulu*, *lizulu*, *nezulu*, and words containing *zulwini*, *tixo*, *kristu*, *yesu*, *msindisi*, *ngcwele* (bare *zulu* is excluded: *kwa zulu* is Zululand). Property/political cues: words containing *taitile*, *mteto*, *rulumente*, *palamente*, *mantyi*, *fama*, *voti*, *nyulo*, *rafu*, *lokishi*. Most uses have neither cue. Example lines for a speaker to check are in outputs/land_kwic.md.

Paper	Cue	Uses	Share	95% CI
Imvo	religious	1408	0.9%	0.4–1.4%
Imvo	property/political	1408	26.8%	23.8–29.7%
Isigidimi	religious	353	3.4%	1.7–5.3%
Isigidimi	property/political	353	12.7%	8.5–17.7%



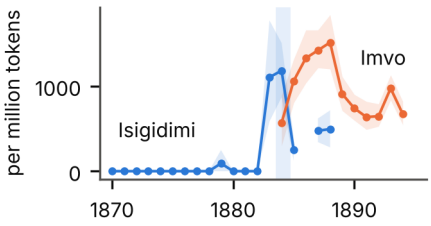


Figure S1 (left): profile-shift percentile of each concept among matched sets (markers as in Fig. 1, left); 'too rare' = too few occurrences, 'seed word' = anchor source, untestable. **Figure S2** (right): parliamentary vocabulary per million tokens; it enters Isigidimi in 1883 (light blue: twice-monthly issues).